$\Rightarrow \tilde{\theta}_3$ наиб. эфф

(Т3)

$$p(x) = \begin{cases} e^{-x/\theta}/\theta, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$\tilde{\theta}_1 = \overline{X}$, $\tilde{\theta}_2 = X_{(2)}$

• $M\tilde{\theta}_1 = M\overline{X} = M\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n} M\left(\sum_{i=1}^{n} x_i\right) = \left\{\begin{matrix}\text{т.к. незав} \\ \text{один. распр}\end{matrix}\right\}$

$= \frac{1}{n}\sum M x_i = \frac{1}{n}\cdot(n \cdot M\xi) = M\xi = \theta \Rightarrow$

несмещенная

• $\varphi(y) = n(n-1)(1-e^{-y/\theta})(e^{-y/\theta})^{n-2} \cdot \frac{1}{\theta} e^{-y/\theta}$

$M\tilde{\theta}_2 = \int_0^{+\infty} y\varphi(y)dy = \int_0^{+\infty} y \frac{n(n-1)}{\theta}\left(e^{-\frac{n+1}{\theta}y} - e^{-\frac{n}{\theta}y}\right)dy$

$$= \frac{n(n-1)}{\theta}\int_0^{+\infty} y e^{-\frac{n+1}{\theta}y} - e^{-\frac{n}{\theta}y}) = \frac{n(n-1)}{\theta}\left(\int_0^{+\infty} y e^{-\frac{n+1}{\theta}y}dy - \int_0^{+\infty} e^{-\frac{n}{\theta}y}dy\right)$$

$$= \frac{n(n-1)}{\theta}\int_0^{+\infty} y e^{-\frac{n+1}{\theta}y}dy - \frac{n(n-1)}{\theta}\int_0^{+\infty} e^{-\frac{n}{\theta}y}dy =$$

$$= \frac{n(n-1)}{\theta}\left(\frac{\theta^2}{(n-1)^2} - \frac{\theta^2}{n^2}\right) = \frac{\theta n(n-1)}{n^2(n-1)^2}(n^2-n^2+2n-1) =$$

$$\theta \cdot \frac{2n-1}{n(n-1)}$$

$M\tilde{\theta}_2 = \theta\frac{2n-1}{n(n-1)} \neq \theta \Rightarrow$ смещенная

$\tilde{\theta}_2' = \frac{n(n-1)}{2n-1}\tilde{\theta}_2$.

$\tilde{\theta}_2'$ - несмещенная

• Сравним эффективность

$\tilde{\theta}_1$:

$D\tilde{\theta}_1 = D\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n}D\xi = \frac{\theta^2}{n}$

для $n=3$ $D\tilde{\theta}_1 = \frac{\theta^2}{3}$

$\tilde{\theta}_2'$:

$D\tilde{\theta}_2' = \left(\frac{n(n-1)}{2n-1}\right)^2 D(\tilde{\theta}_2) = \theta^2\left(\frac{n(n-1)}{2n-1}\right)^2$

$D\tilde{\theta}_2' = \theta^2 \cdot \frac{2n^2-2n+1}{(2n-1)^2}$

для $n=3$: $D\tilde{\theta}_2' = \frac{13}{25}\theta^2$

$\frac{1}{3} < \frac{13}{25}$

$\frac{1}{3}\theta^2 < \frac{13}{25}\theta \Rightarrow \tilde{\theta}_1$ более эфф.

проверим регулярность:

$p(x)$ непр дифф по θ на $(0, +\infty)$

$$\frac{\partial}{\partial\theta}\underbrace{\left(\int_0^{+\infty}\frac{1}{\theta}\exp\left[-\frac{x}{\theta}\right]dx\right)}_{1} = \frac{\partial}{\partial\theta}(1) = \frac{\partial}{\partial\theta}(const) = 0$$

$$\int_0^{+\infty}\left(\frac{x}{\theta^3}e^{-\frac{x}{\theta}} - \frac{1}{\theta^2}e^{-\frac{x}{\theta}}\right)dx = \frac{1}{\theta} - \frac{1}{\theta}$$

$$\Rightarrow \frac{\partial}{\partial\theta}\int_0^{+\infty}p(x,\theta)dx = \int_0^{+\infty}\frac{\partial}{\partial\theta}p(x;\theta)dx$$

$$\ln\left(\frac{1}{\theta}e^{-\frac{x}{\theta}}\right) = \ln\frac{1}{\theta} + \ln e^{-\frac{x}{\theta}} = -\ln\theta - \frac{x}{\theta}$$

$$I(\theta) = M\left[\left(\frac{\partial\ln(p(x,\theta))}{\partial\theta}\right)^2\right] =$$

$$= \int_0^{+\infty}\left(\frac{x^2}{\theta^2} - \frac{2x}{\theta^3} + \frac{1}{\theta^2}\right)\frac{1}{\theta}e^{-\frac{x}{\theta}}dx = \frac{1}{\theta^2}$$

$I(\theta)$ - непр и >0 на $(0, +\infty)$

$\Rightarrow$ вер. модель регул.

$\tilde{\theta}_1$ несм

θ м регул.

$D\tilde{\theta}_1 = \frac{\theta^2}{n}$ - огр на $\forall$ комп $(0, +\infty) \Rightarrow \tilde{\theta}_1$ - регул.

$\tilde{\theta}_2'$ - несмещ

ВМ регулярна

$D\tilde{\theta}_2' = \theta^2 \left(\frac{2n^2-2n+1}{4n^2-4n+1}\right)$ огр на $\forall$ компакте из

$(0,+\infty)$

$\Rightarrow \tilde{\theta}_2'$ - регул.

$D\tilde{\theta} \geq \frac{1}{nI(\theta)}$

$D\tilde{\theta}_1 \geq \frac{1}{3 \cdot \frac{1}{\theta^2}} = \frac{\theta^2}{3} = D\tilde{\theta}_1$

$\tilde{\theta}_1$ - эфф оценка

$D\tilde{\theta}_2 \neq \frac{\theta^2}{3} \Rightarrow \tilde{\theta}_2$ не эфф.